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CASE REPORT

TOXICOLOGY

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Human Fatality Due to Thallium Poisoning: Autopsy, Microscopy, and Mass Spectrometry Assays*,†

ABSTRACT: Thallium has been responsible for many intoxications since its discovery; however, toxicological profiles for thallium in human fatalities have not been updated recently. Autopsy, microscopic investigations, and toxicological analyses were performed on a married couple who died from thallium sulfate intended homicidal poisoning. The distribution of thallium was established by inductively coupled plasma mass spectrometry with hair samples showing the highest thallium concentration. Electron microscopy revealed a dystrophic condition of hair with disorganized cuticle and atrophy of the hair bulb. Thallium interacts with cells at different levels, with prominent ultrastructural injuries in the mitochondria and endoplasmic reticulum, and high concentration of electron dense granules observed in the cytoplasm and mitochondria of several organs. Alopecia, toxic encephalopathy, and peripheral neuropathy were diagnosed in the victims and suggested to be crucial implications for thallium poisoning. The analytical procedures used in this case are of considerable forensic importance in the diagnosis of thallium poisoning.

KEYWORDS: forensic science, forensic pathology, forensic toxicology, ultrastructural pathology, thallium poisoning, homicide

Thallium (Tl) is a toxic metal, which was accidentally discovered in 1861. Despite the 1973 WHO recommendation against its use, it is still used as a rodenticide and an insecticide in some countries (1). Importantly, increased use in emerging technologies and high-tech industry raises serious concerns about exposure to all living organisms (2). Thallium compounds are responsible for many occupational and accidental poisonings (3,4); in addition, it has been detected in many homicide cases when its high toxicity and colorless, tasteless, and odorless qualities were surreptitiously used (5–7).

Two cases of thallium poisoning of 218 poisoning deaths from Tongji Center for Medicolegal Expertise in Hubei, China, during 1999–2008 were reported in a retrospective study (8). Clinical presentations, autopsy findings, microscopic investigations, and toxicological analyses on the two victims are described herein to update the knowledge on thallium poisoning in humans.

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Case Reports

Histories

In December 2007, a married couple (49-year-old woman and 50-year-old man) felt sudden numbness and tingling in the fingers and soles of feet, progressing to involvement of hands and a stubborn ache in the bottoms of the feet. About 7 days later, hair loss occurred and was completed 1 month later in the woman. Prominent sensory disturbances dominated clinically in the following months characterized by: numbness, allodynia, and hyperesthesia in the extremities. Primary weakness in the limbs and atrophy of the associated musculature also appeared during the advanced stage. By the end of April 2008, both individuals deteriorated with similar clinical features and were hospitalized within 11 days of each other (April 28 for the female and May 8 for the male). The couple was diagnosed with thallotoxicosis confirmed by urine thallium. Clinical chemistry examinations demonstrated hematological changes, hematuria, proteinuria, leukocyturia, etc. (Table 1). They were treated with Prussian blue and hemoperfusion; nevertheless, the male died on May 11, 2008 and the female victim died 10 days later.

On the basis of police investigations, the suspect who allegedly poisoned the couple had mixed a total of 25 g of thallium sulfate with the table salt on two occasions (December 25, 2007 and April 26, 2008). Hence, the manner of death was ruled homicide.

Autopsy and Histopathologic Findings

Forensic autopsies were performed 2 h after death. All procedures were in accordance with the ethical standards of the responsible committee on human experimentation in China.

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TABLE 1—*Results of Clinical chemistry examination.**

Test Items	Female	Male	Reference Values
Urine thallium (μg/mL)	8.8, 1.5	4.3	0
White blood cell (WBC) counts ($\times 10^9$/L)	8.1, 27.3, 21.4, 22.6	15.62	4–10
Neutrophil (NEU) counts ($\times 10^9$/L)	6.46, 23.71, 18.22, 20.18	13.38	2–7
NEU%	80.21, 86.96, 84.99, 89.31	85.7	50–70
Alanine Transaminase (ALT) (U/L)	27, 31, 61, 51	51	0–40
Aspartate Transaminase (AST) (U/L)	20, 20, 26, 26	118	0–40
WBC in urine	+ + +, +	+ + +	–
Red blood cells in urine	+ + + +, ++	+ +	–
Urine protein	+ +, –	+	–

*Each result of the female was provided in a time sequence. Urine thallium was determined on May 4 and May 15. The WBC and NEU counts, NEU% were detected on April 30, May 15, May 16, and May 19. ALT and AST were tested on May 1, May 12, May 15, and May 16. Urinalyses were performed on April 30 and May 6.

FIG. 1—*Alopecia, perioral dermatitis, and stomatitis in the female.*

External examination of the female body disclosed almost bald head, crusted perioral dermatitis, and stomatitis (Fig. 1), as well as transverse white bands (Mees' lines) on the fingernails. The brain weighed 1200 g. Edema, necroses of neurons, proliferation and necrosis of glial cells, focal neuronophagia, and satellitosis were seen in each part of the brain upon microscopic examination. Amyloid corpuscles were deposited in the corpus callosum and hippocampus. The kidneys weighed 254 g together. The proximal convoluted tubules (PCTs) showed edema, tubular casts, limited vacuolization and coagulative necrosis, and autolysis (Fig. 2). The left lung weighed 600 g and right lung 575 g. Multifocal edema, hyaline membrane, and bronchopneumonia were observed. The liver weighed 971 g. Focal necrosis, hydropic degeneration, and macrovesicular steatosis were noted in hepatocytes.

External inspection of the male body showed atrichia, perioral dermatitis, and Mees' lines. The weight of the brain, kidneys, liver, left and right lung was 1450, 227, 1028, 745 and 851 g, respectively. Encephaledema, necrosis of neurons and glial cells, limited coagulative necrosis of PCTs, bronchopneumonia, and spotty necrosis of hepatocytes were found similar to the pathologic lesions noticed in the woman.

No other lethal diseases were present that could have contributed to the deaths. Deaths were attributed to thallium poisoning.

Transmission Electron Microscopy Examination

Specimens of precentral gyrus, postcentral gyrus, basal ganglion, midbrain, pons, medulla oblongata, cerebellum, kidney, liver, heart, lung, brachial plexus and gastrocnemius were trimmed into blocks of $1.0 \times 1.0 \times 1.0$ mm^3 and fixed in 2.5% glutaraldehyde for preparing ultrathin section (50 nm thick) on

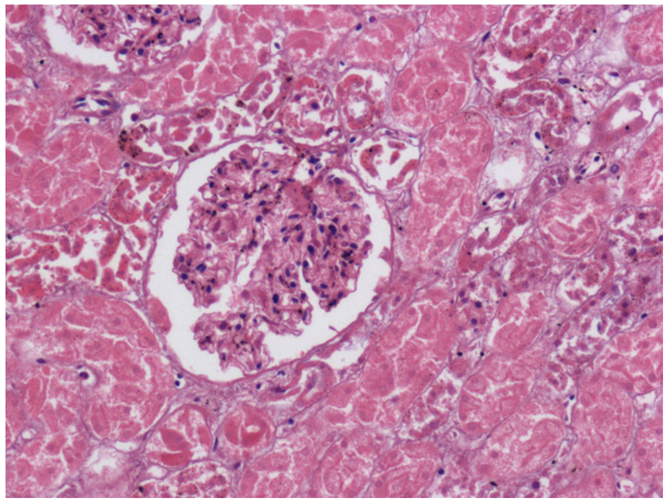

FIG. 2—*Swelling, vacuolization, coagulative necrosis, and autolysis in proximal convoluted tubules (hematoxylin and eosin staining, original magnification ×200).*

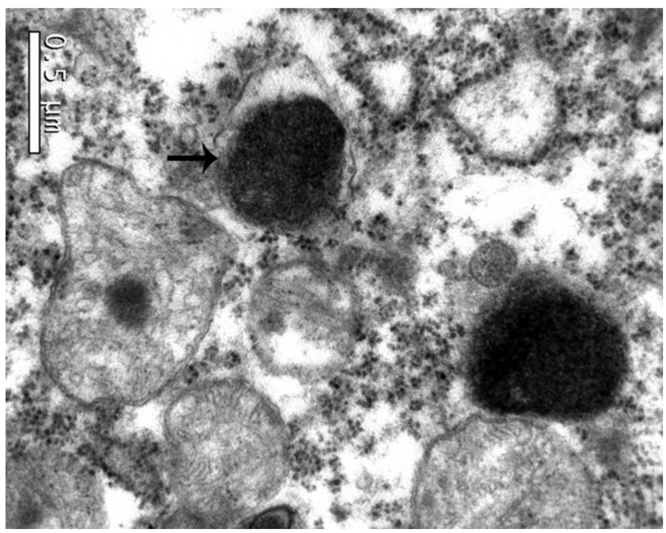

FIG. 3—*Swelling, disorganization, and disintegration in mitochondria of precentral gyrus; high electron dense granules (indicated by black arrow) in mitochondria (Transmission Electron Microscopy).*

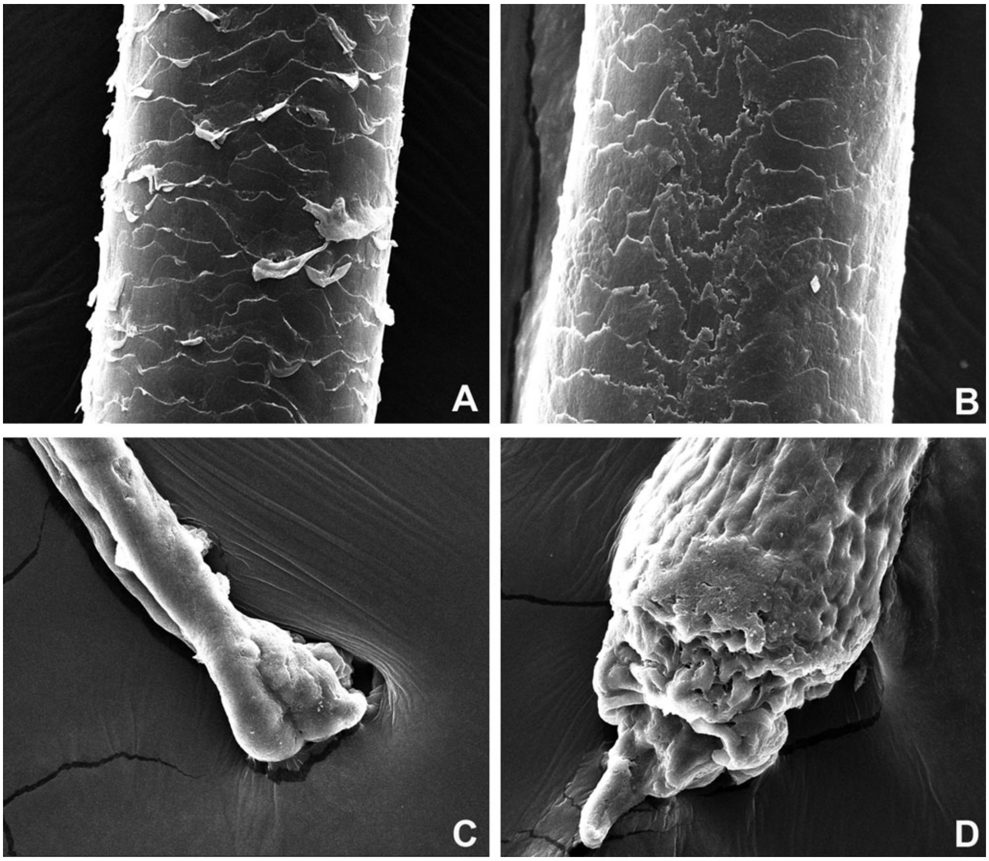

FIG. 4—*Scanning Electron Microscopy images of the hairs. Hair shaft of a victim (A); hair shaft of a healthy individual (B); hair root of a victim (C); hair root of a healthy individual (D).*

an ultramicrotome (PowerTome-PC; RMC Company, St. Louis, Missouri, USA) for Transmission Electron Microscopy (TEM) (Tecnai G2 12; FEI Company, Hillsboro, Oregon, USA) examination.

Universally, significant disruption of the structural integrity of subcellular membrane organelles was observed in neurons, glial cells, hepatocytes, renal tubular epithelial cells, and cardiomyocytes. The most prominent and extensive ultrastructural damage was found in the mitochondria, presenting with marked swelling, conspicuous expansion of intermembrane spaces, and disintegration of membranes and cristae. Swelling and deformed endoplasmic reticulum (ER) with the enlargement of cisternae were disclosed. The number of glycogen granules of smooth endoplasmic reticulum (SER) decreased, while depolymerization of polysomes and degranulation of ribosomes were observed in rough endoplasmic reticulum (RER). High electron dense granules (Fig. 3), likely caused by the deposition of thallium, were found in cytoplasm and mitochondria, especially of the brain, kidney, and liver. These granules were circle-like, measuring about 0.5 μm in diameter. Karyopyknosis, karyolysis, or karyorrhexis were noted, especially in neurons and PCTs. Moreover, the number of euchromatin decreased while heterochromatin increased.

Scanning Electron Microscopy Findings

Hairs collected from the decedents and healthy individual were rinsed with anhydrous acetone for 2 h, dried with a critical point dryer (K850; Quorum/Emitech, East Sussex, U.K.), and then coated with a thin layer of gold using an ion coater (SCD005; Leica EM, Wetzlar, Germany). Thereafter, the hairs were mounted on aluminu stubs and examined by Scanning Electron Microscopy (SEM) (Quanta 200; FEI Company, Hillsboro, Oregon, USA).

A dystrophic condition presenting disorganized cuticle was found in the hair shaft of the victims. The furled and desquamation of cuticle was observed in many locations with multifocal exposure of the cortex (Fig. 4A). On the contrary, that of the healthy individual showed smooth surface and distinct boundaries of cuticular cells (Fig. 4B). The tapered, distorted root and atrophy of the hair bulb were found in the victims (Fig. 4C) compared with the control (Fig. 4D).

Toxicological Analysis

The material preparation and toxicological analysis were performed in Hubei Geological Research Laboratory, China. Specimens, including whole blood from heart, urine, hair, brain, heart, lung, liver, kidney, and stomach, were prepared as previously reported by Das et al. (9) and analyzed in inductively coupled plasma mass spectrometry (ICP/MS) (×7; Thermo Elemental, Waltham, Massachusetts, USA).

The calibration curve was highly linear in the range of 0.1–200 ng/mL ($r^2 = 0.9999$) and defined by the equation, $y = 37854.53x + 1650.65$ (y is the counts per second of ^{205}Tl, and x is the thallium concentration [ng/mL]). The limit of detection and interday precision were 0.003 ng/mL and 4.0–7.9%, respectively. The highest thallium concentration in the hair and the accu-

TABLE 2—*Thallium concentrations in the postmortem specimens of decedents (μg/mL).*

Specimens	Female	Male
Hair	10.20	13.75
Blood	0.019	0.15
Urine	3.15	3.60
Brain	2.03	2.58
Liver	0.98	5.08
Kidney	0.98	2.21
Heart	0.57	1.83
Lung	0.46	0.88
Stomach	0.38	4.02

mulation of thallium in the brain, liver, and kidney are noted in Table 2. In addition, thallium concentrations of the male were much higher than the female, especially in the liver and kidney.

Discussion

According to previous studies, any amount of thallium in the body should be considered abnormal (1,3). Thallium concentrations in normal human and animals should be <1 ng/mL in blood and urine and <10 ng/g in tissues (10). The minimum lethal dose of soluble thallium salts was estimated as 0.7–1.1 g for adults weighing 70 kg or 10–15 mg/kg (11). Thallium is eliminated quite slowly from the body, with the residual thallium detected for months after exposure (12). In animal profiles of acute thallium poisoning, stomach is the first organ showing the maximal concentration, followed by kidney, heart, liver, and brain. It is preferentially accumulated in the kidney, whereas the concentration in brain increases slowly. Thereafter, the brain is the organ showing the largest thallium concentration, whereas thallium decreases in other organs (13). Thallium in the biological matrices of the victims was demonstrated using ICP/MS, which is highly suitable for detection of thallium due to its specificity and sensitivity, accuracy and precision (9). With the longer duration of thallium exposure, but less hospital stays and less effective treatments, the male had higher thallium concentrations. The victims experienced outbreak, chronic poisoning, remission, and acute exacerbation in the histories; these may be attributed to the continuous and small dose intake of tainted salt with thallium sulfate, and the two separate poisonings with an intermission of about 5 months.

The mechanism of thallium toxicity is believed to be interference with vital potassium-dependent processes, as well as high affinity for sulfhydryl groups in proteins and other biomolecules (1,14,15). The metabolic disturbances caused by the inhibitions of enzymes, coenzymes, and structural proteins help to explain the structural changes seen in the cells; such changes appear to correlate with the subsequent physical, neurological, and psychiatric abnormalities in the victims. Based on the TEM studies in these current cases, thallium interacts with cells at different levels. The disruption of membrane integrity of organelles supports an underlying impairment of membrane-bound biological processes that occurred in mitochondria and ER. And the high electron dense granules that possibly resulted from the deposits of thallium, and the changes of chromatin, may indicate some heretofore unrecognized correlations between toxic action and symptomatology from subcellular and genetic levels.

Water-soluble thallium salts have a significant body burden and cross the blood–brain barrier and placenta to affect the brain, heart, kidney, skeletal muscle, testes, and other organs (1,10). Consequently, thallium poisoning is notorious for its complexity and seriousness, often beginning with nonspecific and diverse toxic symptoms (16–18). Alopecia and painful polyneuropathy are its cardinal features, as noted in the two victims here. However, the SEM results of hairs are not enough specific to allow either early diagnosis or chronology of thallium absorption. It was once believed that no significant abnormalities were found in cerebellum, cerebral cortex, basal ganglia, or midbrain from thallium-exposed patients (1); however, the pathologic lesions should not be casually discarded. In our studies, toxic encephalopathy, axonal polyneuropathy, and neuromyopathy were diagnosed in the victims, and these are associated with lesions of sensory and motor neurons, as well as axons (19–21). As to the terminal products (lipofuscin-like pigment or amyloid corpuscle) of lipid peroxidation, they have been closely correlated to oxidative stress and damage caused by a decrease in reductive glutathione (22,23). The victims suffered from lethal respiratory and cardiac disturbances before death, which may be attributed to the most effects of bilateral vagus nerve paralysis leading to development of severe malfunction of cardiac muscle, dyspnea, cyanosis, and death. The gross renal function was not impaired as referred previously (1), whereas mild toxic nephropathy that principally characterized by limited toxic damage in PCTs was confirmed in the victims; thus, instead of dumping protein into urine, mild albuminuria with erythrocytes and leukocytes in the urinary sediment was found.

The wide clinical spectrum of thallium poisoning makes diagnosis difficult especially in the absence of a definite history of exposure. Consistent with the current state of knowledge, thallium poisoning should be considered when alopecia, toxic encephalopathy, and progressive peripheral neuropathy occur. Demonstration of the presence of thallium remains the most important procedure for diagnosing thallium poisoning.

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